

House Price Prediction – Final Project Report

Title: House Price Prediction Report

Dataset: Ames Housing Dataset

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Campaign: DataMites Internship – Advanced Regression Project

1. Problem Statement

Predict the sale price of houses in Ames, Iowa, using various features such as overall quality, living area, and neighborhood characteristics. This project aims to build a robust regression model that helps understand which factors most influence house prices.

2. Objectives

- Perform comprehensive Exploratory Data Analysis (EDA) on the dataset.
- Handle missing values, encoding, and scaling properly.
- Build and evaluate Linear Regression and other machine learning models.
- Identify key features influencing house prices.
- Present a professional, automated workflow suitable for production.

3. Dataset Information

The dataset contains 79 explanatory variables describing various aspects of residential homes.

Target Variable: SalePrice (property's sale price in dollars)

Examples of features:

- OverallQual – Overall material and finish quality
- GrLivArea – Above-ground living area (sq ft)
- YearBuilt – Year the house was originally built
- GarageCars – Garage capacity
- Neighborhood – Location within Ames city limits

Dataset source: Kaggle – House Prices: Advanced Regression Techniques

4. Data Preprocessing

- Loaded and inspected the dataset using Pandas.
- Handled missing values via imputation and column removal when necessary.
- Categorical variables encoded using Label/OneHot Encoding.
- Applied scaling using StandardScaler to normalize numerical features.
- Created custom reusable functions using personal Python library (cwmeda).

5. Exploratory Data Analysis (EDA)

EDA focused on understanding relationships and correlations between variables.

Key Insights:

- OverallQual, GrLivArea, and TotalBsmtSF show strong positive correlation with SalePrice.
- Houses with better materials and more square footage are significantly costlier.
- Few columns (like PoolQC, Fence) had high missing values, dropped after analysis.
- Custom visualizations were created using `m.plot_columns()` and heatmaps.

6. Model Building

A Linear Regression Pipeline was developed including:

- Data encoding
- Feature scaling
- Model training and evaluation

Tree-based models like RandomForestRegressor and XGBoostRegressor were compared for accuracy.

7. Model Evaluation

Model	R ² Score	MSE	RMSE
Linear Regression	0.90	0.01	0.11
Random Forest	0.87	0.017	0.13
XGBoost	0.89	0.15	0.12

Interpretation:

Linear Regression achieved the highest R² score of 0.90, indicating excellent predictive power. Random Forest and XGBoost models also performed well but were slightly lower in R². Linear Regression remains the most interpretable and consistent model.

8. Challenges Faced

1. Missing Values: Columns like PoolQC, Fence, and MiscFeature contained heavy missing data. → Dropped or imputed.
2. Categorical Encoding: Many categorical features required encoding. → Used Label and OneHot Encoding.
3. Outliers: Detected and treated using rule-based detection.
4. Model Overfitting: Solved using StandardScaler and pipeline-based regularization.

9. Conclusions & Learnings

- Developed a full ML workflow: EDA → Preprocessing → Modeling → Evaluation.
- Linear Regression achieved R² = 0.90 on test data.
- Key features influencing house prices identified successfully.

- Learned how to create custom reusable functions, integrate pipelines, and document data projects effectively.

10. Future Scope For Learning Purpose

- Try advanced models like CatBoost or LightGBM.
- Deploy model using Flask/Streamlit.
- Integrate SHAP/LIME for explainable AI.
- Extend cwmeda or my own library for automated insights.

11. Tools & Technologies

Language: Python

Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost

IDE: Jupyter Notebook

Visualization Tools: Custom EDA library (cwmeda)

12. Overall Summary

This project demonstrates complete data analysis and machine learning lifecycle understanding — from data cleaning to prediction. The results are reliable, interpretable, and structured in a production-ready manner.